

Answers.

1. Canonical sum-of-products = $x'y'z' + x'y'z + x'yz + xy'z' + xyz = F(x,y,z)$

Canonical product-of-sum = $(x+y'+z) \cdot (x'+y+z') + (x'+y'+z') = F(x,y,z)$

2. Simplification

$$F(x,y,z) = x'y'z' + x'y'z + x'yz + xy'z' + xyz$$

$$x'y'(z'+z) + x'yz' + yz(x'+x) \quad [\text{distributive}]$$

$$x'y'(1) + x'yz' + yz(1) \quad [\text{inverse}]$$

$$F(x,y,z) = x'y' + yz + x'yz' \quad [\text{identity}]$$

3. Canonical sum-of-products =

$$w'x'y'z' + w'x'y'z + w'x'yz + w'xy'z' + w'xyz + wx'y'z' + wxy'z' + wxyz = F(w,x,y,z)$$

Canonical product-of-sums =

$$(w+x+y'+z) \cdot (w+x'+y+z') \cdot (w+x'+y'+z) \cdot (w'+x+y+z') \cdot (w'+x+y'+z) \cdot$$

$$(w'+x+y'+z') \cdot (w'+x'+y'+z) \cdot (w+x+y+z) = F(w,x,y,z)$$

4. Simplification

$$F(w,x,y,z) = w'x'y'z' + w'x'y'z + w'x'yz + w'xyz + wx'y'z' + wxy'z' + wxyz + w'xy'z'$$

$$= w'x'y'z' + w'x'y'z + w'x'yz + w'xyz + wx'y'z' + wxy'z' + wxyz + w'xy'z' \quad [\text{Associative}]$$

$$= w'x'y'(z'+z) + w'yz(x'+x) + wxy'z' + w'xy'z' \quad [\text{Inverse}]$$

$$F(w,x,y,z) = w'x'y'(1) + w'yz(1) + wxy'z' + w'xy'z' \quad [\text{Identity}]$$

$$\text{SA. } F(x,y,z) = xyz' + xy'z' + x'yz + x'y'z' + xyz$$

$$= xyz' + xy'z' + x'yz + xyz + x'y'z' \quad [\text{Associative}]$$

$$= xz'(y+y') + yz(x'+x) + x'y'z'$$

$$= xz'(1) + yz(1) + x'y'z' \quad [\text{Inverse}]$$

$$F(x,y,z) = xz' + yz + x'y'z' \quad [\text{Identity}]$$

$$\begin{aligned}
B. F(w, x, y, z) &= wx'y'z + wxyz' + wx'y'z' + wx'y'z + w'x'y'z + w'x'y'z' + w'x'y'z + w'x'y'z + w'x'y'z' \\
&= wx'y'z + wxyz' + wx'y'z + wx'y'z' + w'x'y'z + w'x'y'z' + wxyz' + wx'y'z + w'x'y'z' \quad [\text{Associative}] \\
&= wx'y'(z+z') + w'x'y'(z+z') + w'x'y'(z+z') + wxyz' + wx'y'z + w'x'y'z \\
&= wx'y'(1) + w'x'y'(1) + w'x'y'(1) + wxyz' + wx'y'z + w'x'y'z \quad [\text{Inverse}] \\
&= wx'y' + w'x'y' + w'x'y' + wxyz' + wx'y'z + w'x'y'z \quad [\text{Identity}] \\
&= wy'(x+x') + w'x'y + wxyz' + wx'y'z + w'x'y'z \\
&= wy'(1) + w'x'y + wxyz' + wx'y'z + w'x'y'z \quad [\text{Inverse}]
\end{aligned}$$

$$F(w, x, y, z) = wy' + w'x'y + wxyz' + wx'y'z + w'x'y'z \quad [\text{Identity}]$$

$$\begin{aligned}
C. F(w, x, y, z) &= w'x'y' + w'xz + wxz + wx'y'z' \\
&= w'x'y' + xz(w'+w) + wx'y'z' \\
&= w'x'y' + xz(1) + wx'y'z' \quad [\text{Inverse}]
\end{aligned}$$

$$F(w, x, y, z) = w'x'y' + xz + wx'y'z' \quad [\text{Identity}]$$

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